

Measuring the Centre and Spread of a Data Set

Answer Key

Grade 6–7

Quick Answers

1. 1	5. 1, 3	9. 8
2. 2	6. 2	10. 21
3. 2	7. $\frac{4}{11}$	11. 6
4. 5	8. 6	12. $\frac{9}{2}$

Curriculum

Level: Grade 6–7

6.SP.A – *problem 11*

6.SP.B.5 – *problems 1, 2, 3, 4, 5, 6, 8, 9, 10, 12*

7.SP.C.5 – *problem 7*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

3. *If they got 2.5: averaged the six different goal totals on the axis instead of the eleven games*
6. *If they got 5: subtracted the smallest value from the largest, which is the range and not the interquartile range*
9. *If they got 6: reported the middle value of the ordered list, which is the median and not the mean*
12. *If they got 21: used the largest and smallest values, which gives the range rather than the interquartile range*
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What this sheet is teaching, and what to listen for. Two questions run through all twelve problems: *what is typical here?* and *how much do the values differ?* Each has more than one honest answer, and the sheet is built so the answers disagree on the second data set — mean 8 against median 6, range 21 against IQR 4.5. That disagreement is the lesson, not a complication. When a student’s number is wrong, the useful question is almost never “what did you compute?” but “how many values does your number depend on?”

1. The mode. Which number of goals came up most often?

Read the heights of the stacks: 0 goals has two dots, 1 goal has three, 2 goals has two, 3 goals has two, 4 goals has one, 5 goals has one. The tallest stack sits above 1. 1

The mode is a value, not a count. Three games ended 1–nil, but the mode is 1 goal, not 3 games. A student who answers 3 has given the height of the stack instead of the label under it, which is the single most common dot-plot error and worth naming here.

2. The median. What is the median number of goals?

The eleven values in order are 0, 0, 1, 1, 1, 2, 2, 3, 3, 4, 5. With eleven values the middle one is the sixth, because five sit below it and five above.

Counting six dots in from the left lands on a game with 2 goals.

2

Why a dot plot makes this easy. The dots are already in order, so no sorting is needed — just count dots, left to right. Students who write out the list first are not wasting time; they are doing what the plot has already done for them.

3. The mean. What is the mean number of goals per game?

Add every game's goals. Reading stack by stack: $0 + 0 + 1 + 1 + 1 + 2 + 2 + 3 + 3 + 4 + 5$, which is 22 goals in 11 games.

$22 \div 11 = 2$ goals per game.

2

If they answered 2.5: they averaged the six labels on the axis (0, 1, 2, 3, 4, 5) instead of the eleven games. The axis lists the scores that were *possible*; the dots record the scores that actually *happened*, and three of them were 1–nil. Ask how many games their calculation used.

Worth noticing: on this data set the mean and the median are both 2. That is a property of this set, not a rule — problems 8 and 9 are built to break it.

4. The range. What is the range?

The rightmost dot is a game of 5 goals; the leftmost is a game of 0.

$5 - 0 = 5$.

5

The range is a single subtraction, and it uses only two of the eleven games. That is worth saying out loud now, because problem 12 turns it into the whole point.

5. Quartiles. Find Q_1 and Q_3 .

Ordered: 0, 0, 1, 1, 1, **2**, 2, 3, 3, 4, 5. The median (the sixth value) is left out of both halves.

Lower half: 0, 0, 1, 1, 1 — its median is the third of those five, so $Q_1 = 1$.

Upper half: 2, 3, 3, 4, 5 — its median is the third of those five, so $Q_3 = 3$.

Leaving the median out is a convention, and it is the school one. Some books include it. What matters is that the student does the same thing to both halves and can say which convention they used.

6. The interquartile range.

$IQR = Q_3 - Q_1 = 3 - 1 = 2$.

2

In words: the middle half of the team's games — the ones that were not unusually low or unusually high scoring — all fell within 2 goals of each other.

If they answered 5: that is the range, not the IQR. The range asks how far apart the two most extreme games were; the IQR asks how wide the middle half is. Same data, two different questions.

7. One game at random. What is the probability the game had 3 or more goals?

Count the dots at 3 and above: two games of 3 goals, one of 4, one of 5. That is 4 favourable games out of 11 games in total.

$$P = \boxed{\frac{4}{11}}$$

As a decimal that is about 0.36, so a bit better than one game in three.

If they answered $\frac{3}{11}$: they counted the three labels (3, 4 and 5) rather than the four games sitting on them. Every dot on the plot must have an equal chance of being picked, which is only true if you count dots.

8. A different group. Find the median number of books.

The nine totals are already in order: 3, 4, 5, 5, 6, 7, 8, 10, 24. With nine values the middle one is the fifth, with four below and four above.

The fifth value is 6.

6

The 24 does not get a vote here. It sits at the end of the list, so it affects the median only by being one of the four values above the middle — it would make no difference to the answer if Devi had read 240. Hold that thought for problem 11.

9. The mean of the same nine.

$3 + 4 + 5 + 5 + 6 + 7 + 8 + 10 + 24 = 72$ books read by 9 students.

$72 \div 9 = 8$ books each.

8

Look at what has just happened. The mean is 8, but seven of the nine students read fewer than 8 books. A “typical” value that is bigger than most of the data is a warning sign, and it is caused entirely by Devi’s 24.

If they answered 6: they gave the median. Both are measures of centre and both are correct answers to *some* question — but this problem asked for the one that shares the total out equally.

10. The range of the same nine.

Largest 24, smallest 3, so the range is $24 - 3 = 21$ books.

21

One student decides this number. Devi is at one end of the subtraction, so the range describes the gap between her and the least-keen reader, and says nothing at all about the other seven students, who are packed between 4 and 10.

11. Devi is the student who read 24.

(a) Without Devi the eight totals are 3, 4, 5, 5, 6, 7, 8, 10, which add to 48.

$48 \div 8 = 6$ books each.

6

(b) Without Devi there are eight values, so the median is halfway between the fourth and fifth: halfway between 5 and 7, which is 6. The median has not moved at all.

(c) (Open explanation — listen to the reason.)

A model answer. “The median describes this group better. The mean adds every value up, so Devi’s 24 — more than twice anyone else’s total — pulls it up to 8, which is higher than seven of the nine students actually read. Take Devi out and the mean falls all the way to 6, while the median stays at 6 either way. A number that changes that much when one person leaves is not describing the group, it is describing that person. If the question were ‘how many books did the class read altogether?’ the mean would be the right tool, because the mean is the total shared out.”

Marking guide. Full credit for identifying the 24 as the reason the two measures disagree *and* committing to one measure with a reason. Either choice can earn full credit: median because it describes a typical reader, or mean provided the student says the total is what matters. Partial credit for correctly stating both numbers with no account of why they differ. Not sufficient: “the mean is wrong” — the mean is not wrong, it is answering a different question.

If the student is stuck: ask what would happen to each number if Devi had read 240 books. The mean jumps to 32; the median does not move.

Open explanation — see model answer.

12. Two ways to measure spread.

(a) Ordered: 3, 4, 5, 5, **6**, 7, 8, 10, 24, with the median 6 left out of both halves.

Lower half 3, 4, 5, 5: its median is halfway between 4 and 5, so $Q_1 = 4.5$.

Upper half 7, 8, 10, 24: its median is halfway between 8 and 10, so $Q_3 = 9$.

$\text{IQR} = 9 - 4.5 = \boxed{4.5}$

(b) The range, 21, is far larger than the IQR, 4.5.

(c) (Open explanation — listen to the reason.)

A model answer. “The interquartile range gives the fairer picture. The range is worked out from just two values, and here one of them is Devi, so 21 really measures the gap between her and the lowest reader rather than how spread out the group is. The interquartile range measures the width of the middle half of the data, from 4.5 to 9, and Devi cannot change it however many books she reads. Since seven of the nine students are packed between 4 and 10 books, 4.5 describes what this group actually looks like and 21 does not.”

Marking guide. Full credit for saying that the range depends on only the two extreme values (or explicitly on Devi) while the IQR describes the middle half. Partial credit for saying the IQR is better because it ignores the outlier, with no account of what the range measures. Not sufficient: “4.5 is smaller than 21” — true, and not a reason.

A useful follow-up: ask what the range would be if Devi were removed. It drops from 21 to 7, while the IQR of the remaining eight barely shifts. That is the whole argument in one calculation.

Open explanation — see model answer.